

SIMATS ENGINEERING



SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO3 AT1-EXPERIMENTS

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Course Code: CSA0712

Course Name: COMPUTER NETWORK

Part 1: Subnetting & IP Design Experiments

Experiment 2: Subnet Calculation from Host IP

Aim

To calculate the Network Address, First Usable IP, Last Usable IP, and Broadcast Address from a given host IP and subnet mask.

Given

- Host IP: 10.134.164.205
- Subnet Mask: 255.255.255.0
- CIDR: /27

Tools Required

- Wireshark – to observe network packets
- Command Prompt – to check IP configuration
- Calculator – for subnet calculations

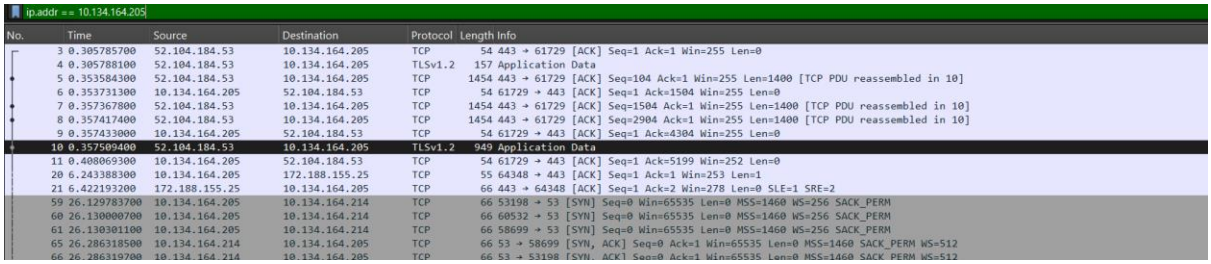
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Wireless LAN adapter Wi-Fi:

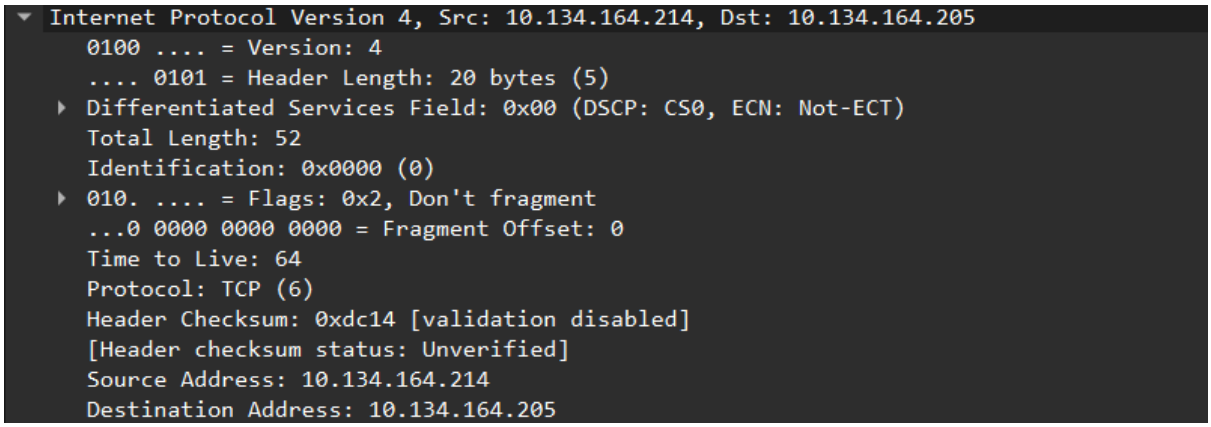
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2401:4900:628c:1201:a212:1a9a:77e4:a2d0
   Temporary IPv6 Address. . . . . . : 2401:4900:628c:1201:9c3:1d1a:ab6b:e79e
   Link-local IPv6 Address . . . . . : fe80::8afa:70f4:fdeb:f4ea%17
   IPv4 Address. . . . . . . . . . . : 10.134.164.205
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::e801:5fff:fe53:3461%17
                                       10.134.164.214

C:\Users\shaik>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=437ms TTL=113
Reply from 8.8.8.8: bytes=32 time=179ms TTL=113
Reply from 8.8.8.8: bytes=32 time=166ms TTL=113
Reply from 8.8.8.8: bytes=32 time=45ms TTL=113

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 45ms, Maximum = 437ms, Average = 206ms
```



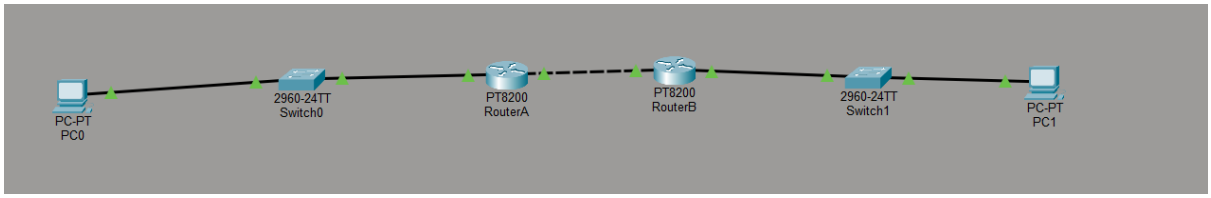


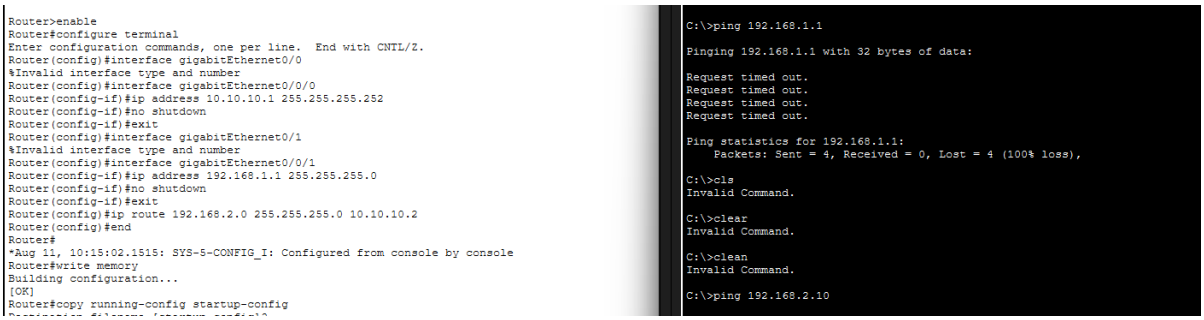
Source IP Address : 10.134.164.214

Destination IP Address: 10.134.164.205

💻 Part 2: Router Configuration & Routing Experiments

Experiment 3: Cisco Command Line Interface (CLI) Static Routing





🔍 Part 3: Wireshark Packet Analysis & Troubleshooting Experiments

Experiment 6: Network Troubleshooting & Path Discovery

Aim

To use tracert to discover the path taken by packets from your computer to 8.8.8.8 and analyze where communication may fail.

Tools

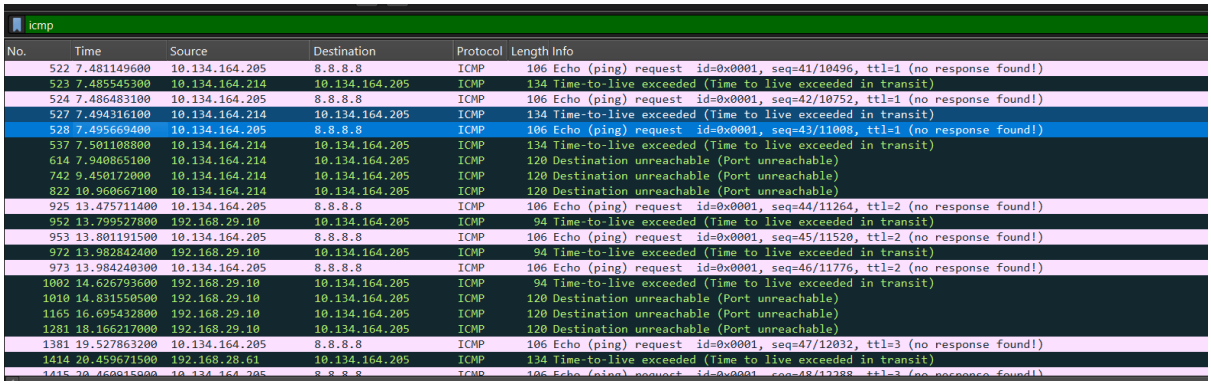
- Cisco Packet Tracer — optional for understanding the topology
- Command Prompt — run traceroute
- Wireshark — capture and analyze packets

```
C:\Users\shaik>tracert 8.8.8.8

Tracing route to dns.google [8.8.8.8]
over a maximum of 30 hops:

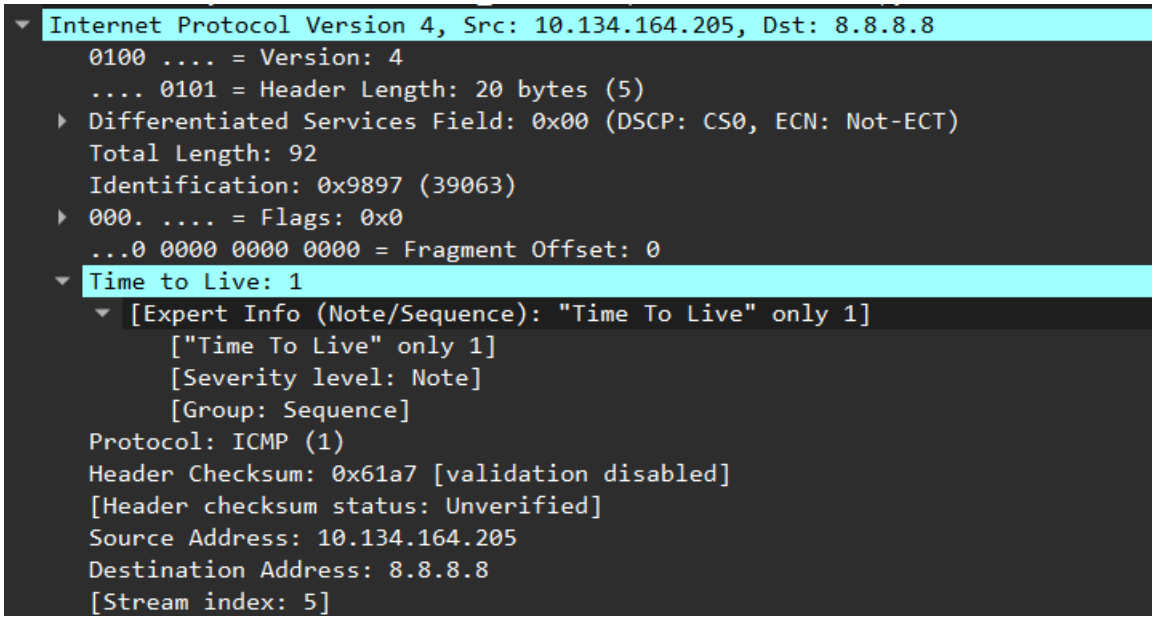
  1     4 ms     7 ms     5 ms  10.134.164.214
  2   324 ms   181 ms   642 ms  192.168.29.10
  3   932 ms   330 ms   118 ms  192.168.28.61
  4   247 ms   233 ms   144 ms  192.168.31.21
  5   374 ms    82 ms   300 ms  192.168.31.33
  6     *        *        *     Request timed out.
  7   377 ms     *        *     nsg-corporate-169.101.187.122.airtel.in [122.187.101.169]
  8     *      812 ms   144 ms  182.79.198.20
  9   164 ms   287 ms   709 ms  142.250.169.206
 10   145 ms   190 ms   121 ms  142.251.66.185
 11   465 ms   298 ms   359 ms  172.253.73.35
 12   224 ms  1353 ms   244 ms  dns.google [8.8.8.8]

Trace complete.
```



icmp

No.	Time	Source	Destination	Protocol	Length	Info
522	7.481149600	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=41/10496, ttl=1 (no response found!)
523	7.485545300	10.134.164.214	10.134.164.205	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
524	7.486483100	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=42/10752, ttl=1 (no response found!)
527	7.494316100	10.134.164.214	10.134.164.205	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
528	7.495669400	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=43/11008, ttl=1 (no response found!)
537	7.501108800	10.134.164.214	10.134.164.205	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
614	7.940865100	10.134.164.214	10.134.164.205	ICMP	120	Destination unreachable (Port unreachable)
742	9.450172000	10.134.164.214	10.134.164.205	ICMP	120	Destination unreachable (Port unreachable)
822	10.960667100	10.134.164.214	10.134.164.205	ICMP	120	Destination unreachable (Port unreachable)
925	13.475711400	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=44/11264, ttl=2 (no response found!)
952	13.799527800	192.168.29.10	10.134.164.205	ICMP	94	Time-to-live exceeded (Time to live exceeded in transit)
953	13.801191500	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=45/11520, ttl=2 (no response found!)
972	13.982842400	192.168.29.10	10.134.164.205	ICMP	94	Time-to-live exceeded (Time to live exceeded in transit)
973	13.984240300	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=46/11776, ttl=2 (no response found!)
1002	14.626793600	192.168.29.10	10.134.164.205	ICMP	94	Time-to-live exceeded (Time to live exceeded in transit)
1010	14.831550500	192.168.29.10	10.134.164.205	ICMP	120	Destination unreachable (Port unreachable)
1165	16.695432800	192.168.29.10	10.134.164.205	ICMP	120	Destination unreachable (Port unreachable)
1281	18.166217000	192.168.29.10	10.134.164.205	ICMP	120	Destination unreachable (Port unreachable)
1381	19.527863200	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=47/12032, ttl=3 (no response found!)
1414	20.459671500	192.168.28.61	10.134.164.205	ICMP	134	Time-to-live exceeded (Time to live exceeded in transit)
1415	20.460915900	10.134.164.205	8.8.8.8	ICMP	106	Echo (ping) request id=0x0001, seq=48/12288, ttl=3 (no response found!)


Internet Protocol Version 4, Src: 10.134.164.205, Dst: 8.8.8.8
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 92
Identification: 0x9897 (39063)
000. .... = Flags: 0x0
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 1
[Expert Info (Note/Sequence): "Time To Live" only 1]
["Time To Live" only 1]
[Severity level: Note]
[Group: Sequence]
Protocol: ICMP (1)
Header Checksum: 0x61a7 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.134.164.205
Destination Address: 8.8.8.8
[Stream index: 5]